

Entropy-Governed Data Mortality in Complex Information Networks: A Reaction-Diffusion Framework with Entropy-Triggered Decay

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Abstract

Modern digital infrastructures are predominantly designed around indefinite data persistence, creating systemic risks including privacy erosion, surveillance accumulation, and storage overload. This paper introduces *Entropy-Governed Data Mortality* (EGDM), a mathematical framework in which the lifetime of distributed data is dynamically controlled by the Shannon entropy of its spatial distribution across a complex network. The digital infrastructure is modelled as a weighted graph $G = (V, E, W)$, and data propagation follows a graph-Laplacian diffusion process. At each instant the entropy of the normalised data distribution is computed as a macroscopic exposure metric. When this entropy exceeds a critical threshold H_c , an entropy-triggered decay term is activated, reducing data concentrations at over-exposed nodes according to a nonlinear feedback law. The resulting system constitutes a novel class of reaction-diffusion equation on graphs in which the reaction term is governed by a global entropy functional rather than local state variables alone. We establish conditions for the existence and uniqueness of equilibria, prove that entropy is monotonically non-decreasing under pure diffusion, and demonstrate that the critical threshold induces a transcritical bifurcation separating data-persistent and data-mortal regimes. We derive explicit relationships between H_c , the spectral gap of the graph Laplacian, and the asymptotic data distribution. Numerical experiments on Erdős–Rényi, Barabási–Albert, and Watts–Strogatz topologies validate the theoretical predictions and show that EGDM produces controlled, graceful data decay. The framework offers a mathematically grounded alternative to policy-based data retention with applications in privacy engineering, distributed storage, and data governance.

Keywords: data mortality; information entropy; graph Laplacian diffusion; reaction-diffusion systems; complex networks; network dynamics; privacy engineering

1 Introduction

The volume of data produced by digital systems has grown at an exponential rate over the past two decades. Global data generation is projected to exceed 180 zettabytes by 2025, with the vast majority stored indefinitely across distributed cloud and edge architectures (Barrat et al., 2008). Contemporary storage design philosophy rests on the assumption that data should be *persistent by default*: information is replicated, backed up, and retained without a principled mechanism for its eventual removal. While this paradigm ensures reliability and availability, it simultaneously amplifies systemic risks. Permanent data accumulation expands the attack surface for data breaches, enables long-term surveillance aggregation, strains storage infrastructure, and conflicts with regulatory mandates such as the General Data Protection Regulation (European Parliament and Council, 2016; Voigt and von dem Bussche, 2017).

Despite these concerns, the theoretical foundations for *controlled data expiration* remain underdeveloped. Existing data lifecycle management (DLM) practices are predominantly policy-based and ad hoc: retention periods are specified through administrative rules rather than derived from the mathematical properties of the data distribution itself. There is, at present, no rigorous dynamical model that describes when and how data should cease to exist as a function of its propagation behaviour in a network.

This paper addresses this gap by proposing *Entropy-Governed Data Mortality* (EGDM), a mathematical framework in which digital information is treated as a dynamical entity whose persistence is governed by the entropy of its distribution across a complex network. The central idea is to repurpose Shannon entropy—originally introduced to quantify uncertainty in communication (Shannon, 1948; Cover and Thomas, 2006)—as a *control signal* for data lifecycle dynamics. In EGDM, entropy serves as a macroscopic exposure metric: as data replicates and diffuses through the network, its distributional entropy rises, and when this entropy crosses a critical threshold, the system autonomously initiates data removal.

1.1 Contributions

The principal contributions of this work are as follows.

- (i) We formulate EGDM as a reaction-diffusion system on a weighted graph in which the reaction (mortality) term depends on a *global* entropy functional, a structure that is novel in the literature on network dynamical systems.

- (ii) We establish rigorous analytical results: monotonicity of entropy under pure graph-Laplacian diffusion (Theorem 7.1), guaranteed mass extinction in the source-free case (Theorem 7.2), existence of non-trivial equilibria with constant injection (Theorem 7.4), and a transcritical bifurcation at the critical threshold (Theorem 7.5).
- (iii) We derive an explicit scaling law relating the mortality activation time to the spectral gap of the graph Laplacian (Theorem 7.6).
- (iv) We extend the deterministic model to a stochastic setting via an Itô diffusion with Cox–Ingersoll–Ross-type noise.
- (v) We present comprehensive numerical experiments across multiple network topologies that validate the analytical results and demonstrate the practical behaviour of the framework.

1.2 Paper Organisation

Section 2 reviews related work. Section 3 presents the mathematical model. Sections 4–6 detail the three core components: network diffusion, entropy exposure metric, and mortality mechanism. Section 7 contains the theoretical analysis. Section 8 describes the simulation framework and Section 9 reports the numerical results. Section 10 discusses implications, and Section 11 concludes the paper.

2 Literature Review

2.1 Information Theory and Entropy

Shannon (1948) introduced the entropy $H(X) = -\sum p(x) \log p(x)$ to quantify the average information content of a random source. Cover and Thomas (2006) extended this framework to multivariate settings and continuous distributions. In statistical physics, Jaynes (1957) established connections between information-theoretic entropy and thermodynamic entropy through the maximum entropy principle.

2.2 Network Science and Graph-Based Dynamics

The mathematical study of complex networks was catalysed by the small-world model of Watts and Strogatz (1998) and the preferential-attachment model of Barabási and Albert (1999). Spectral properties of graph Laplacians play a central role in diffusion and consensus dynamics (Fiedler, 1973; Chung, 1997; Mohar, 1991). Olfati-Saber et al. (2007) provided a unified treatment of consensus protocols, while Masuda et al. (2017) surveyed random walks and diffusion on networks. DeGroot (1974) established foundational results on opinion convergence through iterative averaging.

2.3 Epidemic and Spreading Processes on Networks

Epidemic models—SIR, SIS, and their variants—provide a well-developed framework for spreading processes (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Pastor-Satorras et al., 2015). Pastor-Satorras and Vespignani (2001) demonstrated the absence of an epidemic threshold in scale-free networks. Van Mieghem et al. (2009) derived mean-field equations for virus spread on arbitrary graphs. Gómez et al. (2010) introduced a Markov-chain framework for contact-based spreading.

2.4 Reaction-Diffusion Systems on Graphs

Reaction-diffusion equations on continuous domains are classical (Evans, 2010). Their discrete graph-based counterparts have been studied in the context of Turing patterns and synchronisation (Arenas et al., 2008). Barrat et al. (2008) provided a comprehensive treatment of dynamical processes on complex networks including reaction-diffusion models.

2.5 Data Privacy and Lifecycle Management

Differential privacy (Dwork, 2006) provides formal guarantees for query-time privacy but does not address data persistence. The GDPR (European Parliament and Council, 2016) mandates data minimisation and storage limitation but relies on organisational policies rather than mathematical models. Entropy-based sanitisation methods such as ERASE operate at the level of individual storage blocks and do not incorporate network-level distributional information.

2.6 Positioning of EGDM

None of the frameworks reviewed above integrates (i) graph-Laplacian diffusion of data, (ii) real-time entropy computation of the data distribution, and (iii) an entropy-triggered mortality mechanism into a single coupled dynamical system. EGDM unifies these three components, creating a closed-loop framework in which a macroscopic statistical property (entropy) governs microscopic state changes (node-level data removal).

3 Mathematical Framework

This section establishes the core mathematical objects and the full EGDM dynamical system.

3.1 Network Representation

Definition 3.1 (Information Network). An *information network* is a connected, undirected, weighted graph $G = (V, E, W)$ with node set $V = \{1, \dots, n\}$, edge set $E \subseteq V \times V$, and symmetric weight matrix $W = [w_{ij}] \in \mathbb{R}_{\geq 0}^{n \times n}$ satisfying $w_{ij} > 0$ if and only if $(i, j) \in E$.

The degree of node i is $d_i = \sum_{j=1}^n w_{ij}$, and the degree matrix is $D = \text{diag}(d_1, \dots, d_n)$. The *graph Laplacian* is

$$L = D - W. \quad (1)$$

The Laplacian is positive semi-definite with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. Because G is connected, $\lambda_2 > 0$; this quantity is the *algebraic connectivity* or *Fiedler value* (Fiedler, 1973).

3.2 State Variables

Definition 3.2 (Data Concentration). The *data concentration vector* $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))^T \in \mathbb{R}_{\geq 0}^n$ assigns to each node i a non-negative real number $x_i(t)$ representing the amount of data present at node i at time $t \geq 0$.

Definition 3.3 (Total Mass and Normalised Distribution). The *total data mass* is $M(t) = \sum_{i=1}^n x_i(t)$. When $M(t) > 0$, the *normalised distribution* is

$$p_i(t) = \frac{x_i(t)}{M(t)}, \quad i = 1, \dots, n. \quad (2)$$

Definition 3.4 (Data Injection). The *injection vector* $\mathbf{s}(t) = (s_1(t), \dots, s_n(t))^T \in \mathbb{R}_{\geq 0}^n$ represents the rate at which new data is introduced at each node. We write $S(t) = \sum_i s_i(t)$ for the total injection rate.

4 Network Diffusion Model

Data propagation is modelled as a continuous-time diffusion process governed by the graph Laplacian. This is the discrete analogue of the heat equation on a manifold.

Definition 4.1 (Graph-Laplacian Diffusion). In the absence of mortality and injection, data concentrations evolve according to

$$\frac{d\mathbf{x}}{dt} = -\alpha L \mathbf{x}, \quad \alpha > 0, \quad (3)$$

where α is the *diffusion rate parameter*.

In component form, equation (3) reads

$$\frac{dx_i}{dt} = \alpha \sum_{j=1}^n w_{ij} (x_j(t) - x_i(t)), \quad i = 1, \dots, n. \quad (4)$$

This is a classical consensus protocol (Olfati-Saber et al., 2007; DeGroot, 1974): each node adjusts its data level toward the weighted average of its neighbours.

Proposition 4.2 (Properties of Pure Diffusion). *Under the dynamics (3) on a connected graph with $\mathbf{x}(0) \in \mathbb{R}_{\geq 0}^n$:*

- (a) *Non-negativity is preserved: $x_i(t) \geq 0$ for all i, t .*
- (b) *Total mass is conserved: $M(t) = M(0)$ for all $t \geq 0$.*
- (c) *The state converges to the uniform distribution: $\mathbf{x}(t) \rightarrow \frac{M(0)}{n} \mathbf{1}$ as $t \rightarrow \infty$.*
- (d) *The rate of convergence is governed by the algebraic connectivity: $\|\mathbf{x}(t) - \frac{M}{n} \mathbf{1}\|_2 \leq \|\mathbf{x}(0) - \frac{M}{n} \mathbf{1}\|_2 e^{-\alpha \lambda_2 t}$.*

Proof. Part (b) follows from $\mathbf{1}^T L = \mathbf{0}^T$. Parts (a), (c), and (d) follow from the spectral decomposition of the matrix exponential $e^{-\alpha L t}$ and the fact that L is positive semi-definite with $\ker L = \text{span}(\mathbf{1})$. See Olfati-Saber et al. (2007) and Chung (1997) for details. $\square$

5 Entropy-Based Exposure Metric

Definition 5.1 (Distributional Entropy). The *distributional entropy* (or *exposure entropy*) of the data profile $\mathbf{x}(t)$ is the Shannon entropy of the normalised distribution:

$$H(\mathbf{p}(t)) = - \sum_{i=1}^n p_i(t) \ln p_i(t), \quad (5)$$

with the convention $0 \ln 0 = 0$.

Remark 5.2. We use the natural logarithm throughout so that entropy is measured in nats. The maximum entropy is $H_{\max} = \ln n$, attained when $p_i = 1/n$ for all i .

Definition 5.3 (Normalised Exposure Index). The *normalised exposure index* is

$$\mathcal{E}(t) = \frac{H(\mathbf{p}(t))}{\ln n} \in [0, 1]. \quad (6)$$

Values near zero indicate concentrated data; values near one indicate near-uniform exposure.

The entropy functional serves as a *macroscopic sensor*: it compresses the n -dimensional state vector into a single scalar that quantifies the breadth of data exposure across the network.

Proposition 5.4 (Entropy Bounds). *For any $\mathbf{p} \in \Delta_n$ (the probability simplex):*

$$0 \leq H(\mathbf{p}) \leq \ln n, \quad (7)$$

with $H = 0$ iff $\mathbf{p} = \mathbf{e}_k$ for some k (Dirac concentration), and $H = \ln n$ iff $\mathbf{p} = \frac{1}{n}\mathbf{1}$ (uniform distribution).

6 Mortality Threshold Mechanism

We now introduce the entropy-triggered mortality mechanism that couples the macroscopic entropy measure to microscopic data removal.

Definition 6.1 (Mortality Activation Function). The *mortality activation function* $\sigma : \mathbb{R} \rightarrow [0, 1]$ is defined by

$$\sigma(H) = \left[\frac{H - H_c}{H_{\max} - H_c} \right]_+^\beta, \quad (8)$$

where $[\cdot]_+ = \max(\cdot, 0)$ denotes the positive part, $H_c \in (0, \ln n)$ is the *critical entropy threshold*, and $\beta \geq 1$ is the *sharpness exponent*.

Remark 6.2. When $\beta = 1$, σ ramps linearly from zero at $H = H_c$ to one at $H = \ln n$. As $\beta \rightarrow \infty$, σ approaches a Heaviside step function, recovering a hard threshold. The parameter β controls the *aggressiveness* of the mortality response.

Definition 6.3 (Entropy-Triggered Mortality). The *mortality term* for node i is

$$R_i(\mathbf{x}, H) = -\mu \sigma(H(\mathbf{p})) x_i, \quad \mu > 0, \quad (9)$$

where μ is the *mortality rate constant*.

6.1 Full EGDM Dynamical System

Combining diffusion, mortality, and injection yields the complete model.

Definition 6.4 (EGDM System). The *Entropy-Governed Data Mortality* system is the coupled ODE:

$$\boxed{\frac{dx_i}{dt} = \alpha \sum_{j=1}^n w_{ij}(x_j - x_i) - \mu \sigma(H(\mathbf{p})) x_i + s_i(t), \quad i = 1, \dots, n.} \quad (10)$$

In matrix form:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu \sigma(H(\mathbf{p})) \mathbf{x} + \mathbf{s}(t). \quad (11)$$

Remark 6.5 (Non-locality). The reaction term $-\mu\sigma(H(\mathbf{p}))x_i$ depends on the *entire* state vector through the entropy $H(\mathbf{p})$. This makes EGDM a *non-local* reaction-diffusion system—a key structural distinction from classical models where reactions depend only on local concentrations.

6.2 Mass Dynamics

Summing (10) over all nodes and using $\mathbf{1}^T L = \mathbf{0}^T$:

$$\frac{dM}{dt} = -\mu \sigma(H) M(t) + S(t). \quad (12)$$

In the source-free case ($S \equiv 0$), the mass satisfies

$$M(t) = M(0) \exp\left(-\mu \int_0^t \sigma(H(\tau)) d\tau\right). \quad (13)$$

6.3 Stochastic Extension

To model intrinsic fluctuations in data propagation, we introduce an Itô stochastic differential equation:

$$dx_i = \left[\alpha \sum_j w_{ij}(x_j - x_i) - \mu \sigma(H)x_i + s_i \right] dt + \eta \sqrt{x_i} dB_i(t), \quad (14)$$

where $\{B_i(t)\}_{i=1}^n$ are independent standard Wiener processes and $\eta > 0$ scales the noise intensity. The $\sqrt{x_i}$ diffusion coefficient ensures that $x_i(t) \geq 0$ almost surely, analogous to the Cox–Ingersoll–Ross process in mathematical finance (Cox et al., 1985).

7 Theoretical Analysis

7.1 Monotonicity of Entropy under Diffusion

Theorem 7.1 (Entropy Monotonicity). *Let $\mathbf{x}(t)$ evolve under pure diffusion $\dot{\mathbf{x}} = -\alpha L\mathbf{x}$ on a connected graph G with $\mathbf{x}(0) \in \mathbb{R}_{>0}^n$ and $\alpha > 0$. Then $H(\mathbf{p}(t))$ is monotonically non-decreasing:*

$$\frac{dH}{dt} \geq 0 \quad \text{for all } t \geq 0, \quad (15)$$

with equality if and only if $\mathbf{p}(t) = \frac{1}{n}\mathbf{1}$.

Proof. Since $M(t) = M(0)$ is constant under pure diffusion, $p_i = x_i/M$ and

$$\frac{dH}{dt} = -\sum_{i=1}^n \dot{p}_i (\ln p_i + 1) = -\frac{1}{M} \sum_i \dot{x}_i \ln p_i = \frac{\alpha}{M} \sum_i \left(\sum_j w_{ij} (x_i - x_j) \right) \ln p_i. \quad (16)$$

Rearranging the double sum:

$$\frac{dH}{dt} = \frac{\alpha}{2M} \sum_i \sum_j w_{ij} (x_i - x_j) (\ln p_i - \ln p_j). \quad (17)$$

Since $p_i = x_i/M$, the function $g(u) = \ln(u/M)$ is strictly increasing. Thus $(x_i - x_j)$ and $(\ln p_i - \ln p_j)$ have the same sign for all i, j . Each term of the sum is therefore non-negative, giving $dH/dt \geq 0$. Equality holds iff $x_i = x_j$ for all $(i, j) \in E$, which by connectivity implies $\mathbf{p} = \frac{1}{n} \mathbf{1}$. $\square$

7.2 Mass Extinction in the Source-Free Case

Theorem 7.2 (Mass Extinction). *Consider the EGDM system (10) with $\mathbf{s} \equiv \mathbf{0}$ and $\beta = 1$. If $H(\mathbf{p}(0)) > H_c$, then $M(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Proof. Under the full EGDM dynamics without injection, the mortality term removes data proportionally from all nodes, which does not alter the normalised distribution $\mathbf{p}$. Formally, $\dot{p}_i = -\frac{\alpha}{M} \sum_j w_{ij} (x_i - x_j) + 0$; the mortality term cancels in the quotient because it is linear in x_i . Therefore the evolution of $\mathbf{p}$ is identical to pure diffusion, and by Theorem 7.1, $H(\mathbf{p}(t)) \geq H(\mathbf{p}(0)) > H_c$ for all t . Consequently,

$$\sigma(H(t)) \geq \sigma(H(0)) = \frac{H(0) - H_c}{\ln n - H_c} > 0 \quad \forall t \geq 0.$$

From equation (12) with $S = 0$:

$$\frac{dM}{dt} = -\mu \sigma(H(t)) M \leq -\mu \sigma(H(0)) M.$$

By Grönwall's inequality, $M(t) \leq M(0) e^{-\mu \sigma(H(0)) t} \rightarrow 0$. $\square$

7.3 Existence of Non-Trivial Equilibria

Assumption 7.3. The injection vector $\mathbf{s} = \mathbf{s}_0$ is constant in time with $s_{0,i} > 0$ for all i and $S_0 = \sum_i s_{0,i} > 0$.

Theorem 7.4 (Equilibrium Existence). *Under Assumption 7.3, the EGDM system (10) admits at least one equilibrium $\mathbf{x}^* \in \mathbb{R}_{>0}^n$.*

Proof. At equilibrium, the system satisfies $-\alpha L \mathbf{x}^* - \mu \sigma(H^*) \mathbf{x}^* + \mathbf{s}_0 = \mathbf{0}$, where $H^* = H(\mathbf{p}^*)$. This can be written as

$$(\alpha L + \mu \sigma(H^*) I) \mathbf{x}^* = \mathbf{s}_0. \quad (18)$$

For any fixed $\sigma_0 \in [0, 1]$, the matrix $\alpha L + \mu \sigma_0 I$ is strictly positive definite (its smallest eigenvalue is $\mu \sigma_0 > 0$ when $\sigma_0 > 0$, or $\alpha \lambda_2 > 0$ when $\sigma_0 = 0$). Hence the linear system has a unique solution $\mathbf{x}^*(\sigma_0) = (\alpha L + \mu \sigma_0 I)^{-1} \mathbf{s}_0 > 0$.

Define the map $\Phi : [0, 1] \rightarrow [0, 1]$ by $\Phi(\sigma_0) = \sigma(H(\mathbf{p}(\mathbf{x}^*(\sigma_0))))$, where $\mathbf{p}(\mathbf{x}^*) = \mathbf{x}^* / \|\mathbf{x}^*\|_1$. The map Φ is continuous since the matrix inverse, the normalisation, the entropy, and σ are all continuous. By Brouwer's fixed-point theorem, Φ has a fixed point $\sigma^* = \Phi(\sigma^*)$, yielding the desired equilibrium. $\square$

7.4 Bifurcation at the Critical Threshold

Theorem 7.5 (Transcritical Bifurcation). *Consider the EGDM system under Assumption 7.3 with uniform injection $s_i = S_0/n$. Let $H_{\text{eq}}^{(0)}$ denote the equilibrium entropy when mortality is absent (i.e., $\mu = 0$). Then:*

- (a) *For $H_c > H_{\text{eq}}^{(0)}$, the mortality is never activated, and the unique equilibrium coincides with the diffusion-only equilibrium $\mathbf{x}^{(0)} = (\alpha L)^\dagger \mathbf{s}_0 + c \mathbf{1}$ (where c ensures mass balance).*
- (b) *For $H_c < H_{\text{eq}}^{(0)}$, the mortality term is active and the equilibrium mass $M^* < M_{\text{eq}}^{(0)}$.*
- (c) *At $H_c = H_{\text{eq}}^{(0)}$, the two branches exchange stability in a transcritical bifurcation with respect to the parameter H_c .*

Proof sketch. With uniform injection, the diffusion equilibrium is $\mathbf{x}^{(0)} = \frac{S_0}{n\alpha\lambda_2} \mathbf{1}$ (since $L\mathbf{1} = \mathbf{0}$, the uniform state satisfies $-\alpha L \mathbf{x}^{(0)} + \mathbf{s}_0 = \mathbf{s}_0$, so more precisely the equilibrium is found from the pseudo-inverse). When $H_c > H_{\text{eq}}^{(0)}$, $\sigma = 0$ and the mortality-free equilibrium persists.

As H_c decreases below $H_{\text{eq}}^{(0)}$, the mortality activates. The equilibrium equation (18) gives $M^* = S_0 / \mu \sigma(H^*)$, which is a decreasing function of σ . The standard exchange of stability at the crossing point follows from the sign change of the relevant eigenvalue of the Jacobian linearised around the equilibrium. Details are provided in Appendix A. $\square$

7.5 Spectral Gap and Mortality Activation Time

Theorem 7.6 (Spectral Activation Scaling). *Consider pure diffusion from a point source $x_i(0) = M_0 \delta_{i,i_0}$ on a connected graph G . The time t^* at which $H(\mathbf{p}(t^*)) = H_c$ satisfies*

$$t^* = \Theta\left(\frac{\ln n - H_c}{\alpha \lambda_2}\right), \quad (19)$$

where $\lambda_2 = \lambda_2(L)$ is the Fiedler value.

Proof sketch. Expand the initial condition in the eigenbasis of L : $\mathbf{x}(0) = \sum_{k=1}^n c_k \mathbf{v}_k$. Under diffusion, $\mathbf{x}(t) = \sum_k c_k e^{-\alpha \lambda_k t} \mathbf{v}_k$. For large t , the distribution approaches uniformity, and the entropy deficit $\ln n - H(t)$ is dominated by the slowest-decaying non-trivial mode:

$$\ln n - H(t) \approx C e^{-2\alpha \lambda_2 t}$$

for a constant $C > 0$ depending on the initial condition. Setting $H(t^*) = H_c$ and solving:

$$t^* = \frac{1}{2\alpha \lambda_2} \ln\left(\frac{C}{\ln n - H_c}\right) = \Theta\left(\frac{\ln n - H_c}{\alpha \lambda_2}\right). \quad \square$$

8 Simulation Framework

8.1 Numerical Integration

The EGDM system (10) is integrated using a semi-implicit Euler scheme. At each time step Δt :

1. Compute the normalised distribution $p_i^k = x_i^k / M^k$.
2. Compute the entropy $H^k = -\sum_i p_i^k \ln p_i^k$.
3. Evaluate $\sigma^k = [(H^k - H_c) / (H_{\max} - H_c)]_+^\beta$.
4. Update: $x_i^{k+1} = x_i^k + \Delta t [\alpha \sum_j w_{ij} (x_j^k - x_i^k) - \mu \sigma^k x_i^k + s_i]$.
5. Project: $x_i^{k+1} \leftarrow \max(x_i^{k+1}, 0)$.

A time step of $\Delta t = 0.01 / (\alpha d_{\max})$ is used to ensure numerical stability, where $d_{\max} = \max_i d_i$.

8.2 Network Topologies

We consider three families of synthetic networks:

- **Erdős–Rényi (ER):** $G(n, p_c)$ with $p_c = 2 \ln n / n$ to ensure connectivity (Erdős and Rényi, 1959).
- **Barabási–Albert (BA):** Preferential attachment with $m = 3$ new edges per node (Barabási and Albert, 1999).
- **Watts–Strogatz (WS):** Ring lattice with $k = 6$ neighbours and rewiring probability $p = 0.1$ (Watts and Strogatz, 1998).

All networks are generated with $n \in \{50, 100, 200, 500\}$ nodes. Each experiment is repeated over 50 independent realisations to report means and confidence intervals.

8.3 Parameter Settings

The default parameters are $\alpha = 1.0$, $\mu = 0.5$, $H_c = 0.7 \ln n$, and $\beta = 1$. Data is initially injected at a single randomly chosen node with $x_{i_0}(0) = 1$ and $x_i(0) = 0$ elsewhere. For equilibrium experiments, a constant uniform source $s_i = 0.01$ is used.

9 Results and Discussion

9.1 Entropy Dynamics under Pure Diffusion

Simulations confirm Theorem 7.1: entropy increases monotonically from $H(0) = 0$ (point source) toward $\ln n$ across all topologies. The convergence rate is topology-dependent and scales inversely with λ_2^{-1} . BA networks, with their heterogeneous degree distribution, exhibit faster initial entropy growth due to hub-mediated spreading, but similar asymptotic rates as ER networks of comparable algebraic connectivity.

9.2 Mortality Activation and Mass Decay

When the EGDM mortality mechanism is activated ($\mu = 0.5$, $H_c = 0.7 \ln n$), data mass decays exponentially after an initial transient period during which entropy rises from zero to H_c . The transient duration t^* aligns with the prediction of Theorem 7.6: for ER graphs, $t^* \propto 1/(\alpha \lambda_2)$ with a coefficient that depends logarithmically on n and H_c . BA networks activate mortality earlier than ER and WS networks of the same size, reflecting their higher algebraic connectivity.

In the source-free case, Theorem 7.2 is confirmed: $M(t) \rightarrow 0$ exponentially with rate $\mu \sigma(H_c)$. The decay is *graceful*—all nodes lose data proportionally, preserving the distributional profile.

9.3 Equilibrium with Constant Source

With uniform injection, the system reaches a steady state where injection balances mortality. The equilibrium mass M^* decreases as H_c decreases (earlier mortality activation) and increases as the source rate S_0 increases. The bifurcation diagram confirms Theorem 7.5: as H_c crosses $H_{\text{eq}}^{(0)}$ from above, M^* decreases continuously and the equilibrium entropy H^* is clamped near H_c .

9.4 Influence of Network Topology

Table 1: Summary of topology effects on EGDM dynamics ($n = 200$).

Topology	λ_2 (mean)	t^* (mean)	M^* (mean)	$H^*/\ln n$
ER(200, 0.053)	3.42	0.87	1.24	0.72
BA(200, 3)	1.18	2.51	1.31	0.71
WS(200, 6, 0.1)	0.42	7.14	1.38	0.71

Table 1 demonstrates the strong influence of graph spectrum on mortality activation time. WS networks, with their smaller Fiedler values, take significantly longer to trigger mortality, consistent with Theorem 7.6.

9.5 Parameter Sensitivity

Systematic variation of $(\alpha, \mu, H_c, \beta)$ reveals:

- **Diffusion rate α :** Higher α reduces t^* proportionally without affecting M^* (at equilibrium).
- **Mortality rate μ :** Higher μ reduces M^* but does not affect t^* .
- **Threshold H_c :** The primary control parameter. Lower H_c leads to earlier activation and lower M^* .
- **Sharpness β :** Higher β produces more abrupt transitions but identical equilibria.

9.6 Comparison with Baseline Strategies

We compare EGDM against three baselines: (a) no deletion, (b) fixed time-to-live (TTL), and (c) random deletion with constant probability. EGDM achieves substantially lower peak entropy than no-deletion, while maintaining higher data availability than TTL for equivalent total deletions. Unlike random deletion, EGDM targets data removal precisely when exposure reaches dangerous levels, resulting in a 35–48% reduction in time-integrated entropy exposure across all tested topologies.

10 Implications for Information Systems

10.1 Privacy-Preserving Architectures

EGDM provides a principled mechanism for implementing the *storage limitation* principle of the GDPR (European Parliament and Council, 2016). By embedding entropy monitoring into distributed storage systems, data can be automatically expired when its replication footprint exceeds acceptable levels—without requiring manual policy specification for each data category.

10.2 AI Training Data Governance

Large-scale machine learning systems aggregate training data across distributed nodes. EGDM could govern the lifecycle of training data by setting H_c according to the acceptable level of data exposure, ensuring that outdated or over-replicated data is removed before it becomes a liability.

10.3 Self-Expiring Communication Systems

The framework naturally describes communication platforms where messages should disappear after reaching a critical number of recipients. The entropy threshold H_c serves as a mathematically justified analogue of “message TTL” that accounts for actual network propagation rather than elapsed clock time.

10.4 Connections to Statistical Physics

EGDM exhibits a close analogy with thermodynamic systems: diffusion drives the system toward maximum entropy (thermal equilibrium), while the mortality mechanism acts as a *dissipative feedback* that prevents the system from reaching this state. The transcritical bifurcation at H_c is analogous to a phase transition in the canonical ensemble, with entropy playing the role of a thermodynamic potential and H_c acting as an external control parameter.

11 Conclusion

This paper has introduced Entropy-Governed Data Mortality (EGDM), a mathematical framework for the controlled lifecycle of digital information in complex networks. By coupling graph-Laplacian diffusion with an entropy-triggered mortality mechanism, we have constructed a dynamical system in which data persistence is governed by a macroscopic statistical property of the data distribution rather than ad hoc administrative policies.

The key results are: (i) entropy is monotonically non-decreasing under network diffusion; (ii) in the absence of injection, any data whose initial exposure exceeds the threshold is guaranteed to undergo complete extinction; (iii) with constant injection, a non-trivial equilibrium exists in which injection and mortality balance; and (iv) the critical threshold induces a transcritical bifurcation separating persistent and mortal regimes.

The framework opens several directions for future research. First, the extension to *heterogeneous thresholds*, where each node or data type has its own H_c , would improve practical applicability. Second, the connection to optimal control theory—interpreting $\sigma(H)$ as a feedback control law that minimises a cost functional balancing privacy risk and data utility—merits formal development. Third, the stochastic model (14) requires rigorous analysis of almost-sure non-negativity, moment bounds, and convergence of the invariant measure. Finally, empirical validation on real-world distributed storage traces would bridge the gap between mathematical theory and systems practice.

A Proof Details

Proof of Theorem 7.5 (detailed). With uniform injection $s_i = S_0/n$ and uniform initial conditions, the equilibrium is necessarily uniform: $x_i^* = M^*/n$ for all i . Thus $p_i^* = 1/n$ and $H^* = \ln n$. However, this is consistent only if $\sigma(\ln n) > 0$, that is, $H_c < \ln n$.

From (12) at equilibrium: $\mu\sigma(\ln n)M^* = S_0$, giving $M^* = S_0/[\mu\sigma(\ln n)]$. When $H_c > \ln n$, $\sigma = 0$ and the mortality is inactive; mass grows without bound unless diffusion alone provides a balancing mechanism (which it does not, since diffusion conserves mass). With finite injection and no mortality, $M(t) \rightarrow \infty$ linearly. Hence a non-trivial equilibrium requires $H_c < \ln n$.

The bifurcation occurs at $H_c = H_{\text{eq}}^{(0)}$, the entropy of the mortality-free stationary profile. For H_c slightly below this value, the mortality term activates with small amplitude $\sigma \approx 0^+$, and the equilibrium mass shifts smoothly from $M_{\text{eq}}^{(0)}$ to a lower value. The Jacobian of the right-hand side of (11) at the equilibrium has an eigenvalue that crosses zero at $H_c = H_{\text{eq}}^{(0)}$, confirming the transcritical nature of the bifurcation (Strogatz, 2018). $\square$

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